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# SGDAgent: Hierarchical Spatio-Temporal Scene Graph-Driven Agent for Structured Long-Form Video Understanding

(/pdf?id=ROMVxQfDsk)

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Conference, Area Chairs, Senior Program Committee, Program Committee, Authors
Revisions (/revisions?id=ROMVxQfDsk) BibTeX
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**TL;DR:** We propose the Scene Graph-Driven Agent (SGDAgent), which co-designs a temporally hierarchical scene-graph video index and adaptive multi-granularity retrieval for structured long-video reasoning, achieving performance beyond strong baselines.

**Abstract:**
Long-form video understanding must locate sparse evidence and reason over long-range spatio-temporal interactions under lossy sampling and limited context. Recent agentic approaches build video captions into searchable indexes for iterative querying, but flat text and embeddings provide limited support for structured retrieval across temporal scales. In this work, we propose the \textbf{S}cene \textbf{G}raph-\textbf{D}riven Agent (SGDAgent), which co-designs a temporally hierarchical scene-graph video index and adaptive multi-granularity retrieval for structured long-video reasoning. The index links frame-level graphs, segment-level graphs, and a global event graph through cross-level temporal containment, while a shared entity registry maintains cross-scale identity. Corresponding retrieval tools let the agent adapt its search granularity before targeted frame inspection. Across four benchmarks, SGDAgent remains competitive with baselines using stronger reasoning models. Under matched LVBench settings, adding hierarchical graph retrieval channels to the caption-only baseline raises end-to-end accuracy from 54.8% to 64.4%, and the three control strategies further raise it to 73.0%. Agent trajectory analysis further illustrates how graph-anchored candidate verification restructures the search process.

**Primary Topic:** CV: Visual Reasoning & Symbolic Representations
**Secondary Topics:** CV: Language, Vision & Multi-modal
**Country of Institutions:** China
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**Submission Number:** 12350

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## Paper Decision

Decision by Program Chairs 25 Sept 2026, 02:41 (modified: 25 Sept 2026, 18:29)

Program Chairs, Authors, Program Committee Submitted, Area Chairs, Secondary Senior Program Committee, Senior Program Committee

Revisions (/revisions?id=NGc5Nu1ecB)

**Decision:** Reject

**Comment:**

This paper did not advance to Phase 2 review for the AAAI-27 conference. The paper was reviewed by at least two human reviewers and at least one senior member of the program committee in Phase 1. In addition, papers with at least one positive review were examined by one program chair or associate program chair to ensure the rejection was justified. **This decision is final**. The provided reviews contain more detail, which we hope will be beneficial to the progress of this research.

## Review for Paper 12350

Official Review by Program Committee sksr 31 Aug 2026, 17:22 (modified: 07 Oct 2026, 03:21)

Program Chairs, Area Chairs, Senior Program Committee, Program Committee sksr, Authors, Program Committee Submitted

Revisions (/revisions?id=K3hgyuW7hW)

**Paper Summary:**

This paper presents SGDAgent, a long-video reasoning framework that combines a temporally hierarchical scene-graph index with adaptive agentic retrieval and control. The video is represented at three levels—frame-level scene graphs, segment-level graphs, and a global event graph—with a shared entity registry maintaining identity across temporal scales. At inference time, the agent selects among corresponding global-, segment-, and frame-level retrieval tools, with additional caption backoff and raw-frame inspection for candidate discovery and verification.

In my view, the main contribution is not any individual component in isolation—scene-graph extraction, hierarchical temporal representations, graph retrieval, query routing, and visual verification have all been explored in related forms—but rather a system design that combines these components in a specific way for long-video agents. In particular, the paper couples representation granularity with agent search granularity, allowing the agent to move between global, segment, and frame-level evidence and selectively inspect raw frames. The matched-setting experiments suggest that this design improves both reasoning accuracy and search efficiency.

**Strengths:**

1. The paper addresses an important limitation of caption- and embedding-based long-video agents: flat retrieval representations provide limited support for structured search across temporal scales. Mapping different scene-graph levels directly to corresponding retrieval operations provides an intuitive interface between video representation and agent search.

2. The global event graph appears particularly useful for reducing search cost. Adding the global level reduces repeated LLM and frame-inspection calls while improving accuracy, suggesting that hierarchical retrieval can shift computation from repeated local exploration toward coarse candidate proposal followed by selective visual verification.
3. The qualitative trajectory illustrates how the agent performs coarse-to-fine search (GR, SR, FR, FI) and verifies graph-derived candidates against raw frames. Reporting answer rate, failure-to-terminate rate, LLM calls, and frame-inspection calls in addition to accuracy also provides a more complete characterization of the agentic system.

**Weaknesses:**

1. The methodological novelty appears somewhat incremental and primarily system-level. Most individual ingredients—scene-graph extraction, hierarchical temporal representations, entity alignment, graph-based retrieval, coarse-to-fine search, query routing, confidence-based control, and frame-level visual verification—are established ideas. The main novelty therefore seems to lie in coupling these components into a particular long-video agent architecture, especially the correspondence between representation granularity and retrieval granularity. The empirical results indicate that this integration is effective, but I am less convinced that the paper introduces a fundamentally new representation or reasoning mechanism. The paper would benefit from positioning this system-level co-design more explicitly as its central contribution and clarifying how it differs conceptually from prior hierarchical retrieval and graph-based video agents.
2. The fixed three-level hierarchy is heuristic and may not generalize across videos with different temporal structures. SGDAgent assumes a fixed frame>segment>global hierarchy, with segment-level graphs constructed using 30-second windows. It is unclear why exactly three levels are appropriate across videos of substantially different durations and event densities.
3. Frame-, segment-, and global-level graphs are all constructed using VLM-generated representations. Errors at the frame level can therefore propagate into segment-level aggregation and subsequently into global event relations. The raw-frame inspection mechanism provides some protection at inference time, but it does not address errors that prevent the relevant candidate region from being retrieved in the first place. The paper acknowledges the lack of a standard benchmark for hierarchical scene-graph evaluation and reports manual inspection of a small subset, but this provides limited quantitative evidence about graph correctness.

**Justification of Recommendation:**
The paper is well written, the proposed system is intuitive, and the matched-setting ablations provide convincing evidence that hierarchical graph retrieval and agent control improve a caption-based long-video agent. My main reservation concerns the level of methodological novelty and generality. The individual technical ingredients are largely established, and the contribution appears primarily to be a system design that combines them in a specific way for long-video agents. This system-level contribution is empirically meaningful, but the hierarchy itself is manually fixed and the quality and robustness of the VLM-generated scene graphs are not sufficiently characterized.

**Specific Points of Feedback for Rebuttal:**

1. What is the main methodological distinction between SGDAgent and prior systems combining hierarchical video representations with graph-based or coarse-to-fine retrieval? Do the authors view the primary contribution as a new representation, a new agent-control mechanism, or the co-design of the two?
2. How sensitive is SGDAgent to the choice of hierarchy depth and segment duration? For example, does the same three-level hierarchy remain effective for substantially shorter or longer videos?
3. Since each graph level is generated using a VLM, have the authors evaluated how errors in frame-level scene graphs propagate to segment- and global-level graphs and ultimately to QA performance?

**Rating:** 6: Marginally above acceptance threshold
**Confidence:** 4: The reviewer is confident but not absolutely certain that the evaluation is correct

## Review for 12350

Official Review by Program Committee PETs 28 Aug 2026, 21:48 (modified: 07 Oct 2026, 03:21)
Program Chairs, Area Chairs, Senior Program Committee, Program Committee PETs, Authors, Program Committee Submitted
Revisions (/revisions?id=AXE0icPW5T)

**Paper Summary:**

This paper addresses long-form video understanding, which is plagued by three core challenges: information corruption induced by lossy frame sampling, the limited capability of flat text or embeddings to capture long-range spatio-temporal interactions, and difficulties in localizing sparse cross-segment evidence. Most prior agent-driven video question-answering approaches construct flat caption-centric indexes to support iterative retrieval, yet they lack multi-scale structured semantic representations. To mitigate these limitations, the authors present SGDAgent, a scene-graph-driven agent framework. Offline, it builds a three-tier tree-like spatio-temporal hierarchical scene-graph index consisting of frame-level graphs, segment-level graphs and a global-event graph, where temporal-containment links interconnect different hierarchical layers. A shared entity registry is introduced to enforce consistent entity identity across scales. Three dedicated retrieval operators are correspondingly implemented: Frame-level Retrieval (FR), Segment-level Retrieval (SR), and Global-event-graph Retrieval (GR). The agent reasoning loop additionally incorporates Frame Inspection (FI) for visual grounding and Caption Backoff (CB) as a fallback text-based retrieval channel. Moreover, three agent control strategies are designed: Query-type Routing (R), Confidence Gating (CG), and Terminal Convergence (T). The authors conduct evaluations across four benchmarks: LVBench, LongVideoBench, Video-MME Long, and EgoSchema Val-500. Under matched ablation settings on LVBench, incorporating hierarchical graph retrieval elevates end-to-end accuracy from 54.8% to 64.4%. Enabling the full suite of agent control strategies further pushes the end-to-end accuracy to 73.0%.

**Strengths:**

- Prior works implement multi-granularity only via tooling atop flat metadata. By contrast, SGDAgent constructs a natively interconnected three-tier scene-graph index, where offline graph structures, temporal-containment links and a deterministic shared-entity registry map directly to online FR/SR/GR retrieval operators. This tight index-retrieval coupling enables structured coarse-to-fine graph search beyond pure text-embedding retrieval.
- The paper presents comprehensive ablation experiments that validate the contributions of both caption-based indexing and multi-level scene-graph indexing, while disentangling the roles of different agent control strategies. These results provide empirical evidence for interpreting the internal mechanics of multi-component long-video agents.
- Evaluations are conducted across four mainstream benchmarks to sufficiently validate the effectiveness of the approach. Supplementary material further provides per-category performance breakdowns, revealing the most prominent gains on temporal grounding and event-summarization tasks, which further strengthens the persuasiveness of the experimental conclusions.

**Weaknesses:**

- This paper bears a strong resemblance to DVD, which adopts the general paradigm of "multi-granularity video database + adaptive tool invocation by LLM agent". Although this paper claims to adopt the idea of a hierarchical scene graph, it does not make full use of its hierarchical structure during query processing. Instead, it relies on a router to direct queries to retrievals of different granularities, which essentially results in flat-structured querying. If that is the case, does it have any fundamental differences from DVD?
- The descriptions of some key modules are very vague.
  - Equation (5) sums three scoring terms with unit weight, no ablation is provided to justify equal weighting. It is unclear whether performance would degrade significantly if weights change; whether some term dominates final ranking.
  - Query-type routing logic is unclear. The text describes high-level routing rule: count/color/text question go to FI; causal/plot-level use GR; localization questions start from SR. However it is not clarified: is routing implemented via fixed prompt instruction to LLM? Or rule-based parsing of question tags? Where are the boundaries between different question categories? Whether routing relies on dataset-specific annotation tags or on-the-fly LLM classification for input query.
  - Many important hyper-parameters lack sensitivity ablation. Multiple critical hyper-parameters are hand-tuned: entity-alignment weights (0.7,0.2,0.1), merge threshold 0.78; segment sliding window: 30s window / 5s overlap; global-event grouping: max temporal gap 12s, overlap thresholds 0.18/0.22. None of them have ablation experiments. It is unknown whether good performance hinges heavily on these specific values, or whether results generalize to other videos/datasets.
- Cascading error propagation and VLM hallucination risk in hierarchical graph construction are not analyzed. Hierarchical graph construction is fully bottom-up: segment-level graphs aggregate frame-level outputs; global event graph is purely built from textual representations of segment graphs without feeding visual pixels. Errors or hallucination from lower-level frame-graph extraction will propagate upwards to segment and global graph. The paper does not present error-case analysis, does not quantify how often cascading failure occurs, and lacks discussion on VLM hallucination impact on downstream QA performance. This is a non-trivial risk for bottom-up graph indexing.

- Ambiguous definition for $S_{soft}$ scoring term. (1) which embedding model is used to encode graph entities and relations; (2) how structured entity-relation triples are converted into embedding vectors (e.g., whether entity and relation embeddings are computed separately and then composed. if so, a naive averaging of entity and relation embeddings lacks theoretical justification); and (3) what concrete fuzzy string-matching algorithm is employed in the fallback branch, and whether and how the cosine-similarity and fuzzy-matching scores are calibrated or combined.

**Justification of Recommendation:**
The paper proposes a natively interconnected three-tier scene-graph index tightly coupled to FR/SR/GR retrieval operators. This design is conceptually novel and validated by comprehensive ablations across four benchmarks. However, the paper insufficiently articulates its differentiation against the closely related DVD work, leaving the novelty contribution boundary unclear. Additionally, multiple critical implementation details are unspecified: the $S_{soft}$ scoring term (embedding model, triple-to-vector conversion, fuzzy matching algorithm), query-type routing mechanism, equal weighting in Eq. 5, and numerous hand-tuned hyperparameters all lack clarification or ablation; the fully bottom-up graph construction also carries unanalyzed cascading error and VLM hallucination risks. These omissions substantially undermine reproducibility and confidence in the reported gains. If authors provide an explicit novelty comparison against DVD, the missing implementation details, supplementary ablations, and failure-case analysis in rebuttal, I am ready to raise my score.

**Specific Points of Feedback for Rebuttal:**
see weaknesses

**Rating:** 5: Marginally below acceptance threshold
**Confidence:** 4: The reviewer is confident but not absolutely certain that the evaluation is correct

## AI Review

AI Review by AI Reviewer ( ) 27 Jul 2026, 19:59 (modified: 25 Sept 2026, 05:15)
Program Chairs, AI Reviewer, Area Chairs, Senior Program Committee, Authors, Program Committee Submitted
Revisions (/revisions?id=YiTjms0kYH)

**Review:**
**Title:** SGDAgent: Hierarchical Spatio-Temporal Scene Graph-Driven Agent for Structured Long-Form Video Understanding

**Synopsis of the paper**
SGDAgent addresses long-form video question answering by replacing flat caption search with a temporally hierarchical scene-graph index. The index links frame, segment, and global event representations through shared entities and temporal containment. An agent retrieves at matching granularities, backs off to captions, and inspects raw frames before answering. Experiments on several benchmarks and matched ablations examine accuracy, termination, and online tool use.

**Summary of Review**
The paper presents a coherent coupling between hierarchical video representation and coarse-to-fine retrieval for long-video evidence localization. Matched LVBench ablations show that the complete pipeline improves accuracy and convergence over the internal caption-only configuration. However, the experiments do not isolate hierarchy and containment from the additional VLM-generated semantic information and retrieval channels introduced with the graphs. Recent hierarchical and multiscale memory agents also narrow the novelty to SGDAgent's particular scene-graph schema, temporal links, and retrieval interface. Underspecified retrieval and controller operations, together with the absence of direct graph-quality evaluation, further limit attribution of the gains to the proposed mechanism.

**Strengths**

- The representation and retrieval interface have a direct semantic correspondence. Figures 1–2 and Equations (1)–(4) associate frame, segment, and global event representations with progressively scoped retrieval tools, making the intended coarse-to-fine reasoning path conceptually traceable.
- The evaluation separates answer quality from coverage and termination. Table 2 jointly reports end-to-end accuracy, answered accuracy, answer rate, failed-to-terminate rate, and online calls, revealing that controller improvements arise substantially from resolving more trajectories rather than uniformly improving accuracy among answered questions.

- The empirical scope covers complementary long-video settings. Table 1 evaluates LVBench, LongVideoBench, Video-MME Long, and EgoSchema, while the paper appropriately relies on matched LVBench configurations rather than heterogeneous published results for component attribution.

**Weaknesses**

- The central attribution to hierarchical graph structure is not isolated. In Table 2, adding frame, segment, and global graphs simultaneously adds VLM-generated annotations, offline processing, retrieval channels, and cross-level organization; no equal-information flat serialization, containment-removed graph, or independently searchable multilevel control is evaluated. The $54.8$-to-$64.4$ gain therefore establishes that the additional graph-derived channels help, but not that temporal hierarchy or containment causes the improvement.
- The novelty boundary is narrower than the contribution framing indicates. HAVEN already combines global, scene, segment, and entity-level indexing with agentic search (Yin et al., 2026), while WorldMM adaptively retrieves from multiple memory types and temporal scales (Yeo et al., 2026); neither appears in the supplied references, related-work discussion, or comparison table. These antecedents make the explicit scene-graph schema, shared registry, temporal containment, and corresponding retrievers—rather than hierarchical or multigranular agentic retrieval generally—the differentiating elements.
- The matched baseline does not control for agent sophistication. The caption-only `Base` in Table 2 lacks routing, confidence gating, and terminal convergence, whereas the full system includes all three; no caption-only configuration uses analogous controls, and the stronger DVD rows in Table 1 are explicitly cross-setting. Consequently, the headline baseline-to-full-system difference cannot determine how much of the controller gain depends specifically on graph retrieval.
- The evaluation does not directly validate the constructed graphs or evidence retrieval. The Evaluation Metrics section reports only that an unspecified "small subset" was inspected and content was "generally preserved," without entity-identity accuracy, relation fidelity, temporal-link correctness, evidence recall, or localization quality. Downstream multiple-choice accuracy cannot distinguish benefits from accurate graph structure, redundant generated descriptions, or answer-option matching.
- The central retrieval and control procedures cannot be reconstructed from the specification. Equation (5) omits formulas, ranges, and calibration for $S_{\text{lex}}$, $S_{\text{soft}}$, and $S_{\text{temp}}$ despite summing them without normalization; Algorithm 1 also leaves $H$ unchanged when an `Answer` action fails `ConfidenceGate`, although the prose describes redirection to additional retrieval. These operations directly govern the reported retrieval and convergence effects.
- Several component conclusions rely on small margins without uncertainty estimates. Table 2 attributes effects to changes of $1.2$–$2.2$ accuracy points on the 1,549-question LVBench set, but reports neither repeated runs nor paired confidence intervals or significance tests. The robustness of the smaller global-graph and controller effects therefore cannot be assessed.

**Suggestions for Improvement**

- A failure-stage analysis could separate errors arising from frame sampling, entity registration, global-event abstraction, retrieval, frame inspection, and final reasoning. Contrasting these categories with Figure 3's successful trajectory would clarify how errors propagate across graph levels.
- Results stratified by evidence span and question type would test the motivating hypothesis more directly. Local visual-detail, repeated-entity, cross-segment temporal, and causal questions should benefit differently from the three retrieval granularities if the proposed routing is operating as intended.
- An amortized efficiency analysis could include preprocessing calls, tokens, storage, latency, and the number of questions required to offset index construction. Table 2 currently measures online calls only after the reported 7.94-hour LVBench graph build.
- Replacing repeated presentations of the headline gains with a compact construction example would improve technical clarity. Showing how one frame relation is registered, aggregated into a segment relation, and incorporated into a global event would expose the cross-level semantics more concretely than the overview diagrams.

**Additional Details and Minor Issues**

- Table 1 reports $73.9$ percent on EgoSchema Val 500, but single-run raw-count accuracy over 500 items changes in $0.2$-point increments. This conflicts with the stated raw-count rounding protocol.
- Figure 3 retrieves a frame at 35:25–35:30 after SR identifies 23:45–24:15 and depicts that frame beneath an event ending at 24:40. The timestamp or containment link conflicts with the stated scoped retrieval.

- Equation (1) omits the cross-level support or containment mappings from the formal index, although these links define the claimed hierarchy.
- FTT combines empty outputs, runtime failures, and non-convergent trajectories. Separating these causes would distinguish system reliability from reasoning convergence.

**References**

Yeo, W., Kim, K., Yoon, J., & Hwang, S. J. (2026). WorldMM: Dynamic multimodal memory agent for long video reasoning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* (pp. 25599–25609).

Yin, X., Peng, X., Li, X., Xiong, Z., & Lu, Y. (2026). Hierarchical long video understanding with audiovisual entity cohesion and agentic search. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* (pp. 32882–32891).

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